Monte Carlo Simulation of the Two-Dimensional XY Model

Zhaoyang Chu
s4390199

Zoutong Shen
s4879759

Computational Physics 2026 — Project 2 — Group 12

April 29, 2026

Abstract

1 Introduction

1.1 The XY Model and the BKT Transition

1.2 Vortices and Topological Charge

1.3 Observables

2 Methods

2.1 Metropolis Algorithm

2.2 Equilibration and Sampling Protocol

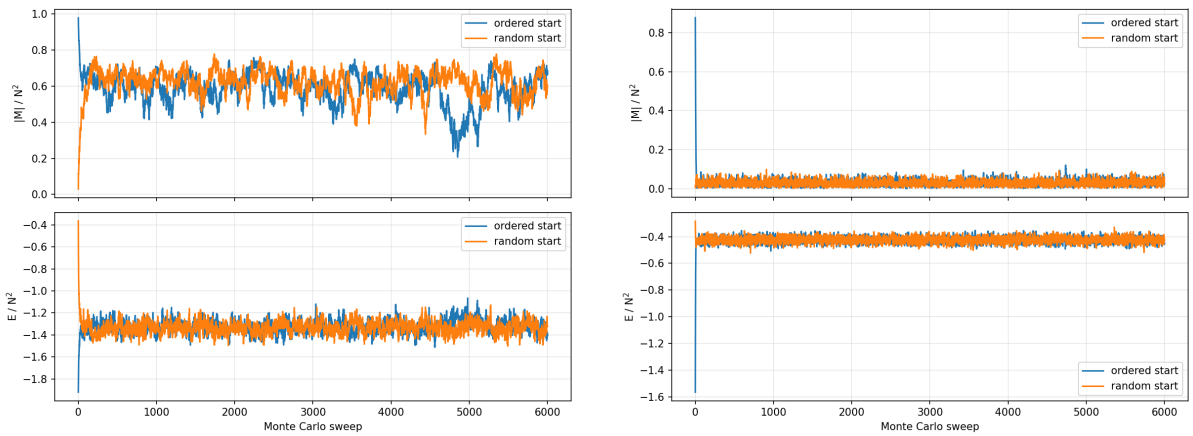
2.3 Error Analysis

2.4 Vortex Detection

3 Results

3.1 Validation

3.1.1 Equilibration: Two Starts



(a) TODO Pass 2: ordered vs random start at $T = 1.0$ (near T_{BKT}), $N = 20$.

(b) TODO Pass 2: ordered vs random start at $T = 2.5$ (deep disordered phase), $N = 50$.

Figure 1: TODO Pass 2: equilibration validation. Two independent starts at the same temperature converge to the same plateau in $|m|$ and e, demonstrating that the simulation is sampling the equilibrium distribution.

3.1.2 Stationarity: Single-run Trace

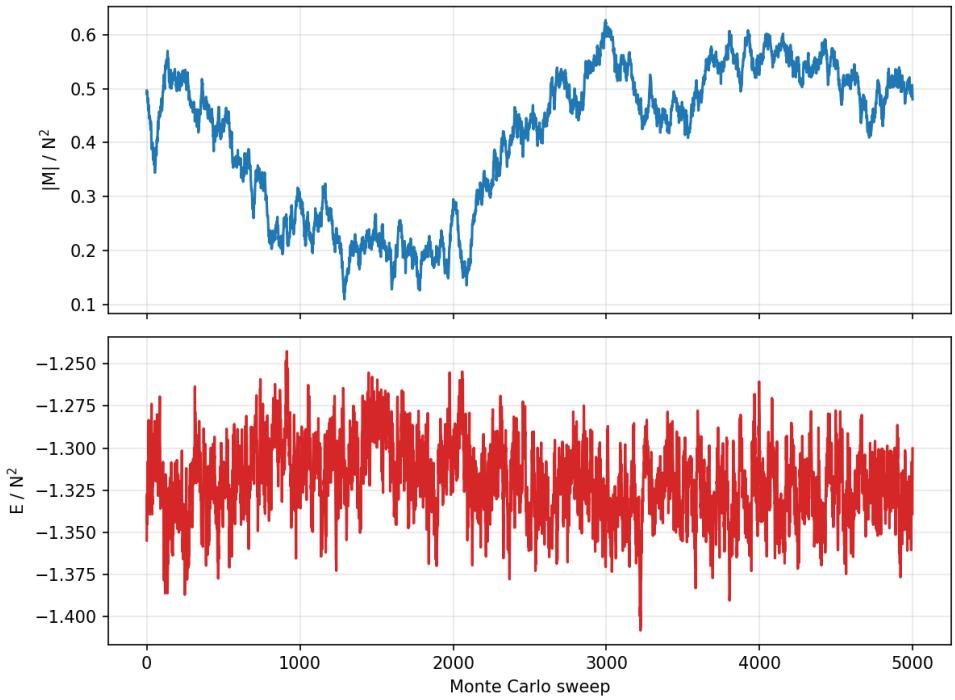


Figure 2: TODO Pass 2: production trace of $|m|$ and e at $T = 1.0$, $N = 50$.

3.1.3 Critical Temperature vs Literature

3.1.4 Finite-size Sanity

3.2 Magnetisation and Susceptibility

3.3 Energy and Heat Capacity

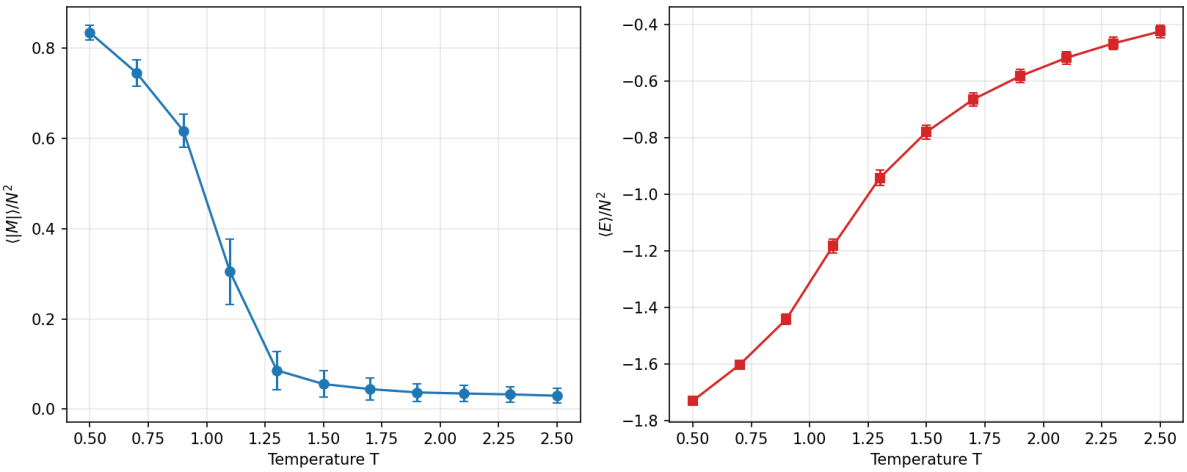


Figure 3: TODO Pass 2: temperature dependence of $\langle|m|\rangle$, $\langle e\rangle$, χ_M, and C for the $N = 50$ scan. Error bars: correlated standard error on the means, blocking method on χ_M and C.

3.4 Correlation Function and Correlation Time

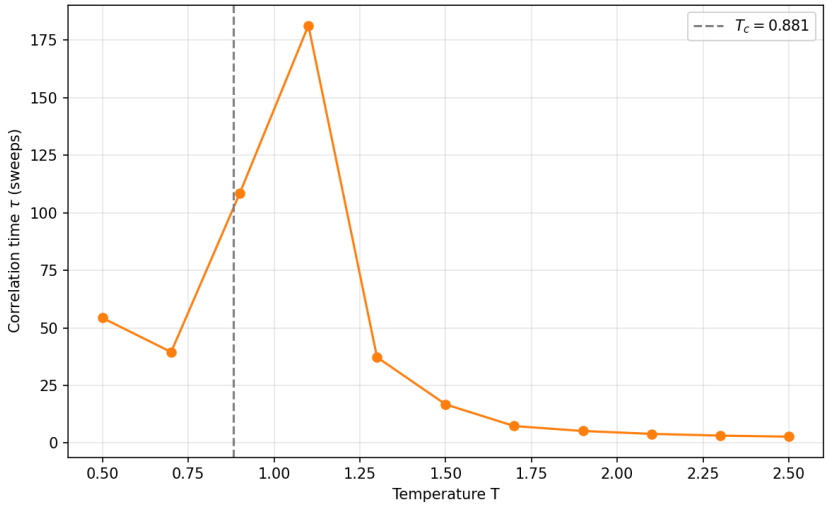


Figure 4: TODO Pass 2: integrated correlation time $\tau(T)$ at $N = 50$ across the 11 scan temperatures (placeholder shows $N = 20$ until regeneration).

3.5 Vortices at Low Temperature

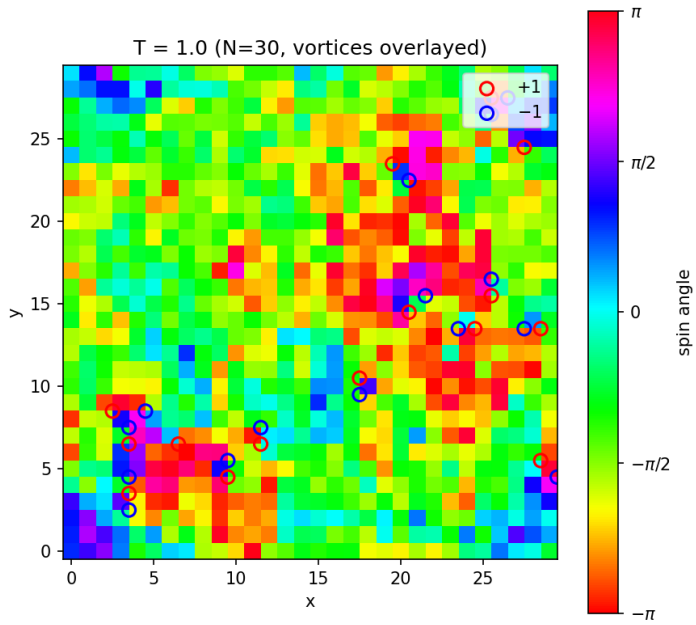


Figure 5: TODO Pass 2: spin configuration at $T = 1.0$, $N = 30$, with $+1$ vortices (red) and -1 anti-vortices (blue) circled. Bound pairs are visible.

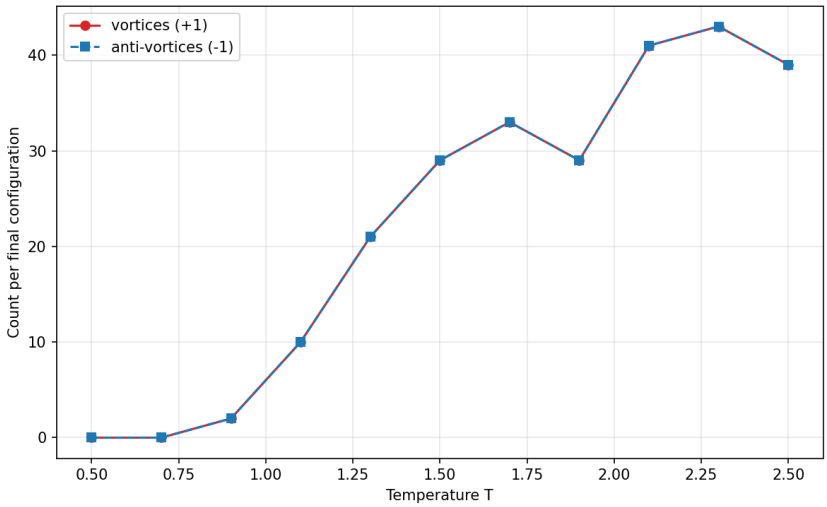


Figure 6: TODO Pass 2: number of $+1$ vortices and -1 anti-vortices in the final configuration as a function of T, $N = 50$ (placeholder shows $N = 20$). On a periodic lattice the two counts coincide.

3.6 Configuration Grid

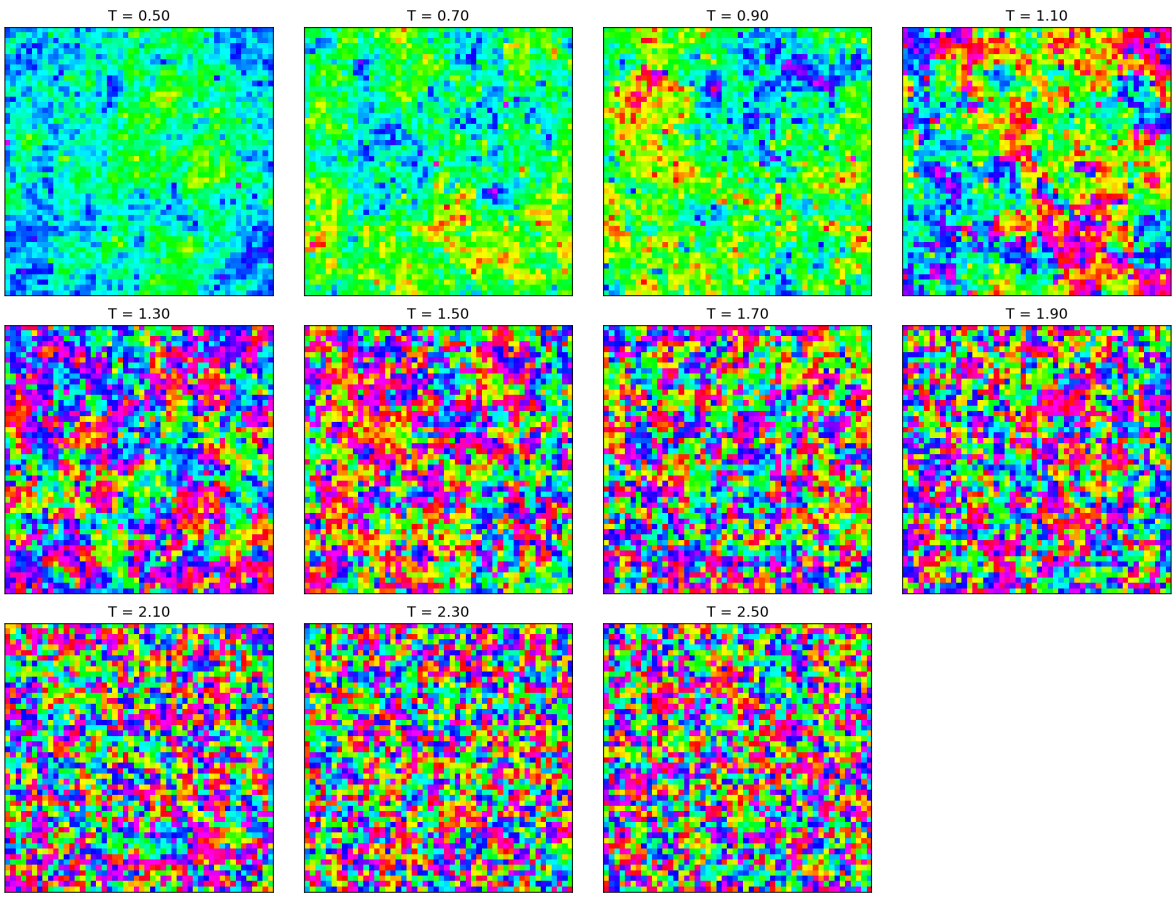


Figure 7: TODO Pass 2: final spin configurations across the temperature scan, $N = 50$, with hue encoding the spin angle.

3.7 Numerical Summary

Table 1: TODO Pass 2: peak quantities from the $N = 50$ temperature scan, with the literature comparison.

Quantity	Value	Source
T at χ_M peak	TBD $\pm$ TBD	this work, $N = 50$
χ_M at peak	TBD $\pm$ TBD	this work, $N = 50$
T at C peak	TBD $\pm$ TBD	this work, $N = 50$
C at peak	TBD $\pm$ TBD	this work, $N = 50$
Literature T_{BKT}	0.8935	Hasenbusch [2005]
Relative deviation of T_χ	TBD %	this work vs literature

4 Extension: Next-Nearest-Neighbour Diagonal Coupling

4.1 Motivation and Hamiltonian

4.2 Method Changes

4.3 Results

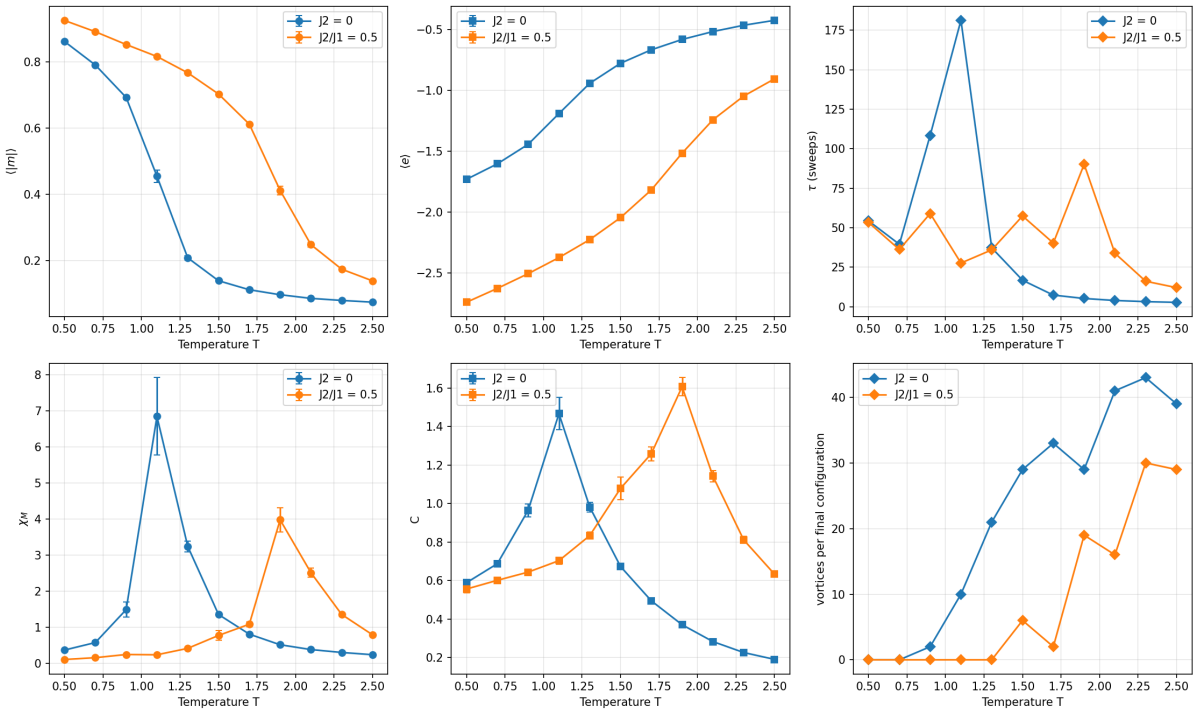


Figure 8: TODO Pass 2: J_1-only vs $J_1 + J_2$ comparison at $N = 20$ for $\langle|m|\rangle$, $\langle e\rangle$, τ, χ_M, C, and the vortex count.

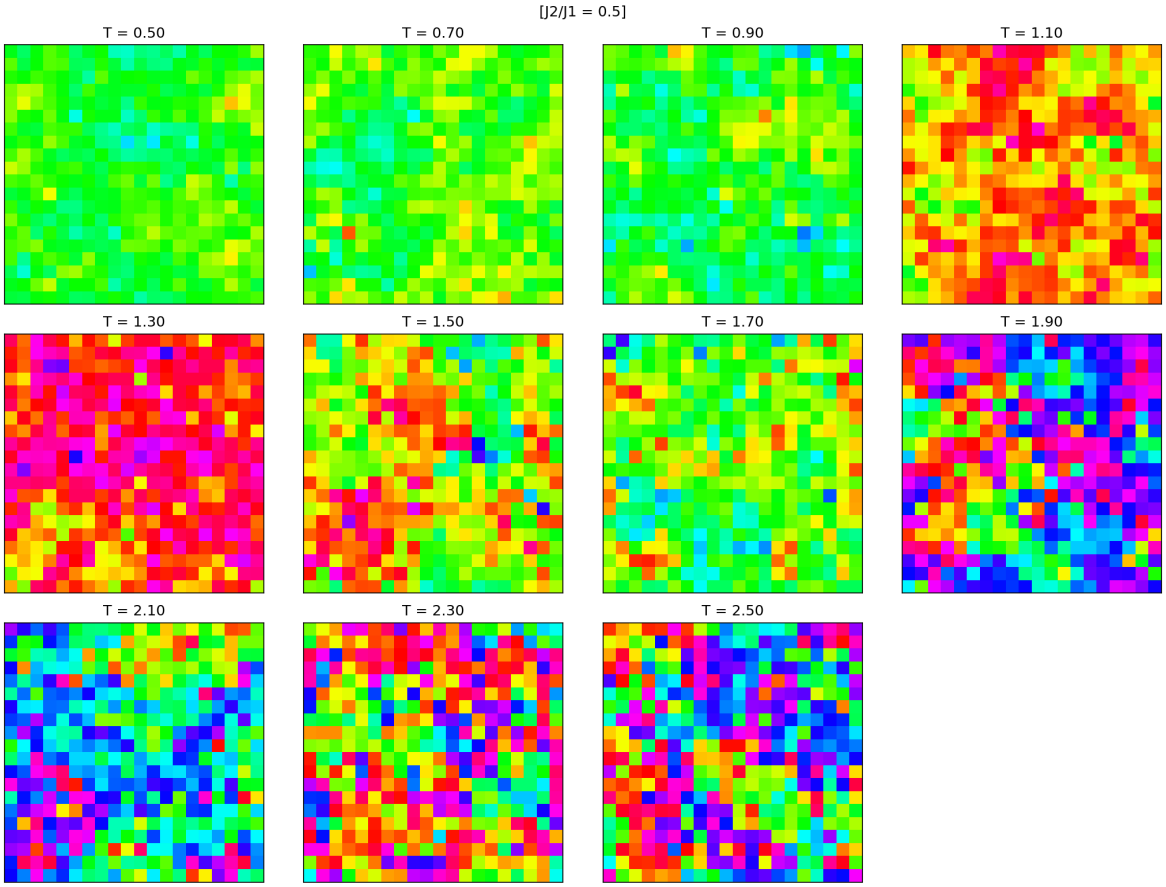


Figure 9: TODO Pass 2: final spin configurations across the temperature scan with $J_2/J_1 = 0.5$, $N = 20$. Smoother than the $J_2 = 0$ case at matched T.

5 Summary

References

M. Hasenbusch. The two-dimensional XY model at the transition temperature: a high-precision Monte Carlo study. *Journal of Physics A: Mathematical and General*, 38(26):5869–5883, 2005. doi: 10.1088/0305-4470/38/26/003.